

		23813 Cortenova (Lecco) ITALY Largo De Vecchi 11 P.o. box 11 Tel. 0341.98341 Fax 0341.983496 - 983497		CERTIFICATO DI COLLAUDO INSPECTION CERTIFICATE CERTIFICAT D'ESSAIS		N. 191621 Rev.01		DATA / DATE 04/04/24		SHEET / 1/1			
		FLANGE E FORGIATI		ORDINE / ORDER / C.DE 1002-PO-013388		EN 10204 3.1							
		DATA / DATE 16/01/24		COMMESSA 75698		FRAMO FLATTOY AS FLATØYVEGEN 24 NO-59 FREKHAUG NORWAY N							
N. RIF.	ITEM	N / QUANTITY	DESCRIZIONE / DESCRIPTION / DÉSIGNATION					NOTE					
SR1852	20	2	F51/F60 PREMACHINED RING					mm Øe473xØi300xH145 A1126					
SR1870	40	2	F51/F60 PREMACHINED RING					QTR Nr. 1 / MDS D44 / MS05 mm Øe473xØi174xH153 A1008					
SR1906	60	2	F51/F60 PREMACHINED RING					QTR Nr. 1 / MDS D44 / MS05 mm Øe471xØi304xH78 A1261 QTR Nr. 1 / MDS D44 / MS05					
N.	MATERIALE / MATERIAL / MATÉRIEL		HEAT / COULÉE		ACCIAIERIA / STEEL PLANT / ACIÉRIE					ORIGIN. CERT. / CERT. D'ORIGIN			
SR1852	ASTM A182 F51/F60		278287		COGNE Refining method A.O.D.					2022049752			
SR1870	ASTM A182 F51/F60		300309		COGNE Refining method A.O.D.					2023008195			
SR1906	ASTM A182 F51/F60		377897		COGNE Refining method A.O.D.					2023053680			
N. RIF.	ANALISI CHIMICA / CHEMICAL ANALYSIS / COMPOSITION CHIMIQUE												
	C	Mn	Si	P	S	Cr	Ni	Mo	N				PRE
Min						21,000	4,500	2,500	0,140				35,000
Max	0,030	2,000	1,000	0,030	0,020	23,000	6,500	3,500	0,200				
SR1852Ladl	0,014	1,180	0,390	0,026	0,000	22,470	5,710	3,240	0,180				36,042
SR1852Ched	0,014	1,200	0,394	0,026	0,000	22,490	5,730	3,250	0,180				36,100
SR1870Ladl	0,021	1,190	0,440	0,020	0,000	22,460	5,760	3,200	0,180				35,900
SR1870Ched	0,021	1,190	0,441	0,020	0,000	22,460	5,760	3,200	0,180				35,900
SR1906Ladl	0,024	1,210	0,370	0,025	0,000	22,460	5,620	3,170	0,180				35,801
SR1906Ched	0,023	1,120	0,352	0,024	0,000	22,370	5,530	3,120	0,177				35,500
N. RIF.	PROVE MECCANICHE / MECHANICAL TESTS / ESSAIS MECANQUES												
	ø mm.	SO mm.2	LU mm.	RS-YIELD LIMITE	R-TENSILE RUPTURE	A. %-ELONG. ALLONGEM.	C. %-REDUCT. STRICTION	RESILLENZA / IMPACT TEST / RESILIENCE				HARDNESS / DURETÉ	
				MPA	MPA	2"		MEDIO / AVERAGE MOYEN	MINIMO MINIMUM	TIPO TYPE	TEMP.	Brinell	
				450,00	655,00	25,00	45,00	Joule	Joule		°C	10-3000-30 Max.271	
A1126TRAN	12,50	122,65	50,00	549,00	739,00	37,48	72,47	159	164	171	KV	- 46	212/214
A1126LONG	12,50	122,65	50,00	516,00	686,00	35,44	71,78	163	172	179	KV	- 46	215/218
A1008TRAN	12,50	122,65	50,00	545,00	705,00	38,17	65,60	157	164	173	KV	- 46	211/215
A1008LONG	12,50	122,65	50,00	488,00	723,00	36,49	67,53	166	175	182	KV	- 46	223/226
A1261TRAN	12,50	122,65	50,00	536,00	715,00	36,66	71,91	162	169	175	KV	- 46	214/216
A1261LONG	12,50	122,65	50,00	586,00	710,00	42,27	79,22	167	176	184	KV	- 46	226/229
N. RIF.	TRATTAMENTI TERMICI / HEAT TREATMENTS / TRAITEMENTS THERMIQUES												
SR1852	Solution Annealed at 1060°C for min 1/2 h/inch quenched water												
SR1870	Solution Annealed at 1060°C for min 1/2 h/inch quenched water												
SR1906	Solution Annealed at 1060°C for min 1/2 h/inch quenched water												
EN 10204 3.1 Surface and dimensional check according to relevant standard. Steel made by electric furnace. Corrosion test acc. to ASTM G48 method "A" at 25°C / 24 hours, no weight loss, no pitting at 20x Ferrite determined using ferritoscope fisher MP30 on all pieces, results: 35-60% on total body. Material acc. to QTR 01 R8; Norsok M630 2020, MDS D44 Rev.6; MDS DH44 Rev.1 as applicable No weld repair. Reduction ratio > 4:1 water temperature before quenching<40°C water temperature after quenching<50°C Mechanical tests according to ASTM A370 Test specimen orientation: transversal and longitudinal Impact test specimen size 10 x 10 x 55 mm The works operate a quality management system acc. to annex I, section 4.3 of the pressure equipment directive (PED) 2014/68/EU certificate by TUV certificate nr 58/2002/MUC. Material acc. to MA-00351 REV F													
ESITO DELLE PROVE / TEST ISSUE / RESULTS DES ESSAIS: FAVOREVOLE SATISFACTORY SATISFAISANT													
TRADE MARK	P. OFF. A. MELESI & C.		COLLAUDATORE / INSPECTOR / INSPECTEUR				ENTE COLLAUDATORE / INSPECTION AGENCY						
	Electronically Generated Certificate		Andrea Gubetti				 AUTORITÉ DE RECETTE						

		23813 Cortenova (Lecco) ITALY Largo De Vecchi, 11 P.o. box 11 Tel 0341 98.341 rie.aut. Fax 0341 90.15.06 - 90.14.88 FLANGE E FORGIATI		CERTIFICATO CONTROLLO CON LIQUIDI PENETRANTI LIQUID PENETRANT TEST			
		Nr.	Data / Date	Pag / Sheet	Commessa / Job		
		24850	04/04/24	1 / 1	75698		
		Ordine / Order		Cliente / Customer			
		1002-PO-013388		1850 FRAMO FLATTOY AS			
Specifica / Specification ASTM A961 S56				Criteri di accettazione / Acceptance criteria ASME VIII Div.1, Appendix 8			
Procedura / Procedure PT 01 - Rev. 1 Ed. 2				Stadio di controllo / Inspection phase After final h.t. machined before drilling			
Pos Item	Q.tà Quantity	Descrizione Description		Colata Heat	Cond. Superficiali Surface status		
20	2	F51/F60 PREMACHINED RING mm Øe473xØi300xH145		278287	FINITO/FINISHED		
40	2	F51/F60 PREMACHINED RING mm Øe473xØi174xH153		300309	FINITO/FINISHED		
60	2	F51/F60 PREMACHINED RING mm Øe471xØi304xH78		377897	FINITO/FINISHED		
TECNICA D'ESAME / EXAMINATION TECHNIQUE							
☐ Fluorescente / Fluorescent ☒ Lavabile con acqua / Water washable ☒ Contrasto colore / Color contrast ☐ Rimovibile con solvente / Solvent removable ☐							
FASI / STEPS		PRODOTTI / PRODUCTS - TIPO / TYPE			LOTTO N. / BATCH NR.		
		FABBRICANTE / MANUFACTURER					
Pulizia / Cleaning		Solvente/Cleaner CGM (R2 60 velnet solnet)					
Applicazione penetrante / Penetrant application		CGM (R2.71 RED "WW")			22/170		
Rimozione eccesso penetrante / Excess penetrant removal		Acqua/Water Pressione/Pressure <275 kPa (40psi) Temperatura/Temperature <40°C					
Asciugatura / Drying		Asciugatura con stracci / Wiping					
Applicazione rilevatore / Developer application		CGM (R.82 ROTRIVEL "U" / white)			23/396		
TECNICA D'ESAME / EXAMINATION TECHNIQUE							
Tempo di penetrazione / Penetrant dwell time		20 min					
Tempo di sviluppo / Developer dwell time		12 min					
ATTREZZATURA DI ILLUMINAZIONE / LIGHTING EQUIPMENT							
BETA 1838SW LAMP							
NOTE / NOTES							
No defects Examination performed on 100% of pieces							
RISULTATI / RESULTS							
Conforme							

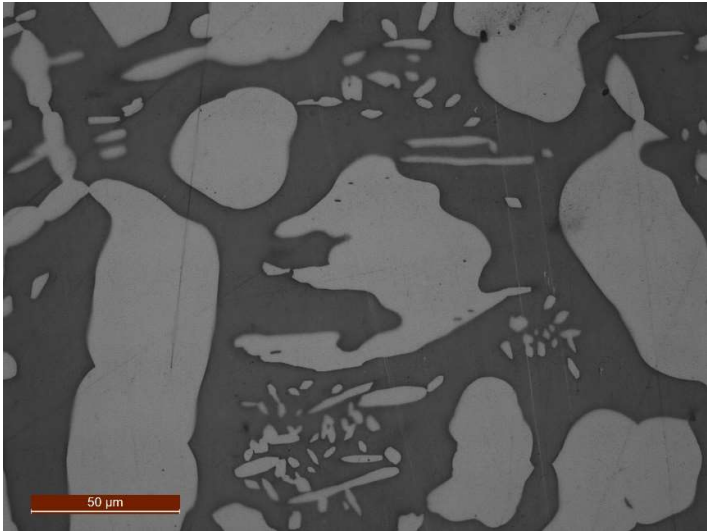
FIRME / SIGNATURES	
OPERATORI / OPERATORS	
LOCATELLI DARIO Lev. II	
ENTE COLLAUDATORE / INSPECTION AUTHORITIES	ISPETTORE CLIENTE / CUSTOMER INSPECTOR

		23813 Cortenova (Lecco) ITALY Largo De Vecchi, 11 P.o. box 11 Tel 0341 98.341 rie.aut. Fax 0341 90.15.06 - 90.14.88 FLANGE E FORGIATI		RAPPORTO ESAME ULTRASUONI ULTRASONIC TEST REPORT			
		Nr.	Data / Date	Pag / Sheet	Commessa / Job		
		24851	04/04/24	1 / 1	75698		
		Ordine / Order		Cliente / Customer			
		1002-PO-013388	1850 FRAMO FLATOY AS				
Specifica / Specification ASTM A388 (+S1)				Criteri di accettazione / Acceptance criteria API 6A PSL3			
Procedura / Procedure UT_005-16 Rev.4				Stadio di controllo / Inspection phase After final h.t. machined before drilling			
Pos Item	Q.tà Quantity	Descrizione Description	Colata Heat	Cond. Superficiali Surface status			
20	2	F51/F60 PREMACHINED RING mm Øe473xØi300xH145	278287	LAVORATO/MACHINED			
40	2	F51/F60 PREMACHINED RING mm Øe473xØi174xH153	300309	LAVORATO/MACHINED			
60	2	F51/F60 PREMACHINED RING mm Øe471xØi304xH78	377897	LAVORATO/MACHINED			
APPARECCHIO / INSTRUMENT							
Tipo / Type		Matricola / Serial Number		Costruita da / Manufactured by			
USN 60		01H3B5		GE			
SONDE / PROBES - Amplificazione durante esame / Amplification during examination +6 dB							
Tipo Type	Costruita da Manufactured by	Matricola S.N.	Misura Size	Angolo Angle	Frequenza Frequency	Ampl. di regolazione Setting ampl.	
SEB2 E	GE Sensing&Inspection Technologies	57467-50712	6 X 20 mm		2 MHz		
B2S	GE Sensing&Inspection Technologies	57745-71949	24 mm		2 MHz		
MWB45-2E	GE Sensing&Inspection Technologies	56921-500041	8 X 9 mm	45°	2 MHz		
CAVI / CABLES							
Tipo Type	Lunghezza Length						
RG 174 D	2 mt						
CALIBRAZIONE / CALIBRATION							
☐ DGS Diagramma DGS Diagram			☒ Blocchi di calibrazione Calibration blocks FBH 1,6-3,2-6,4 mm, notch< 3% T				
☒ DAC			☐ Blocco di riferimento V2 Reference block V2				
☒ Eco di riferimento Reference point echo + 6 db			☐ Blocco di riferimento V1 Reference block V1				
Riflettore di riferimento / Reference reflector FBH 1,6-3,2-6,4 mm, notch< 3% T							
Segnale di riferimento / Reference amplitude FBH 1,6-3,2-6,4 mm, notch< 3% T							
Accoppiante / Coupling Olio / Oil							
NOTE / NOTES							
Conform to specification							
*No single indications exceeding reference distance amplitude curve + No multiple indications exceeding 50% of reference distance amplitude curve							
RISULTATI / RESULTS							
Conforme							

FIRME / SIGNATURES	
OPERATORI / OPERATORS	
MIOTTI SERGIO Lev. II	
ENTE COLLAUDATORE / INSPECTION AUTHORITIES	ISPETTORE CLIENTE / CUSTOMER INSPECTOR

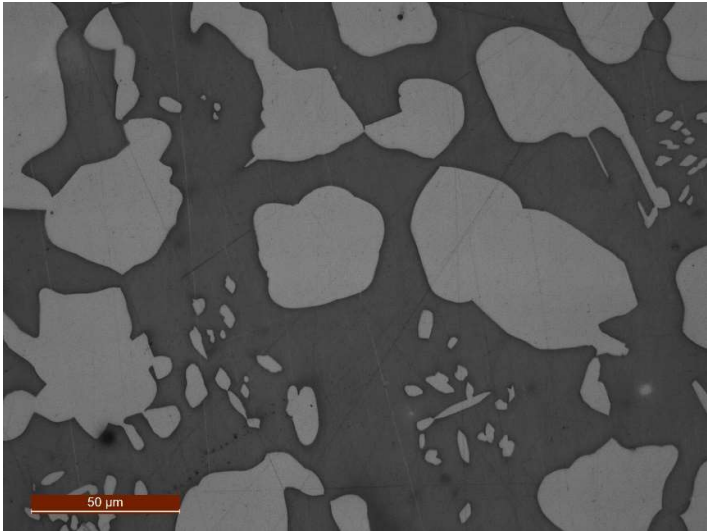
 23813 Cortenova (Lecco) ITALY Largo De Vecchi 11 P.o. box 11 Tel. 0341.98341 Fax 0341.983496 - 983497	DATA / DATE 04/04/2024		CERTIFICATO N° 0867/2024 REPORT N°	
	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	278287	LOTTO LOT	A1126

PROVA DI DETERMINAZIONE DELLE FASI INTERMETALLICHE DANNOSE / DETECTING DETRIMENTAL INTERMETALLIC PHASE TEST

Informazioni generali <i>General Info</i>			
Descrizione prova <i>Test description</i>		Prova di determ.delle fasi interm. dannose: Attacco con sodio idrossido <i>Detecting detrimental intermetallic phase: Sodium Hydroxide Etch Test</i>	
Standard Prova <i>Test standard</i>		ASTM A923-14 Metodo A ASTM A923-14 Method A	
Strumentazione <i>Equipment</i>		Metallographic microscope Leica DMi8 M (OAM n°1631)	
Parametri di prova <i>Test parameters</i>			
Posizione di prelievo <i>Sampling position</i>		Superficie esterna <i>External surface</i>	
Ingrandimento <i>Magnification</i>		500X	
Caratteristiche provino <i>Specimen information</i>			
Preparazione della superficie <i>Surface preparation</i>	Lucidatura 1µm 1µm polishing	Attacco Etching	40% NaOH elettrolitico electrolytic
			
Immagine non in scale / Image not in scale			
☒ Unaffected structure	<i>The ferrite has been etched without revelationi of intermetallic phase The interphase boundaries are smooth</i>		
☐ Possibly Affected Structure	<i>The ferrite has been etched with isolated indications of possible intermetallic phase The interphase boundaries may show a fine waviness</i>		
☐ Affected Structure	<i>The indications of an intermetallic phase are readily revealed before or simultaneously with the staining of the ferrite during etching</i>		
☐ Centerline Structure	<i>An intermetallic phase is observed as a continuous or semi-continuous phase in the mid-thickness region of the product, with or without the affected structure outside of the mid-thickness region, indicative of segregation</i>		
Note / Remarks L'analisi metallografica mostra una struttura regolare libera da fasi intermetalliche e precipitati <i>The results of metallographic examination show a regular structure free from intermetallic phases and precipitates</i>			

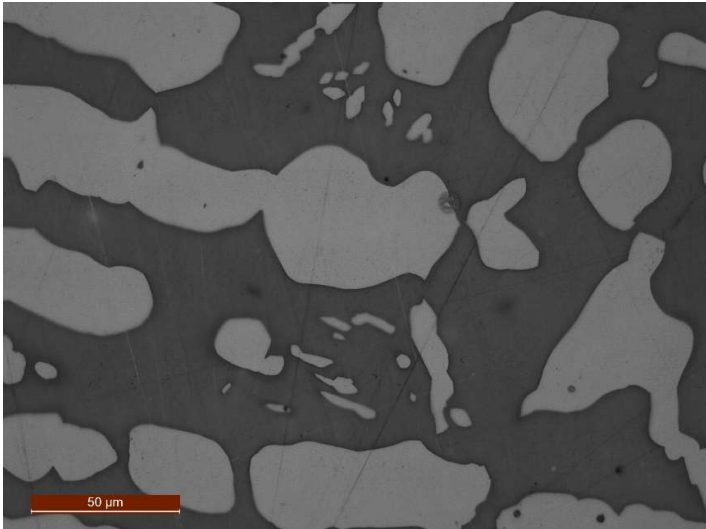
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	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	278287	LOTTO LOT	A1126

PROVA DI DETERMINAZIONE DELLE FASI INTERMETALLICHE DANNOSE / DETECTING DETRIMENTAL INTERMETALLIC PHASE TEST

Informazioni generali General Info			
Descrizione prova Test description		Prova di determ.delle fasi interm. dannose: Attacco con sodio idrossido Detecting detrimental intermetallic phase: Sodium Hydroxide Etch Test	
Standard Prova Test standard		ASTM A923-14 Metodo A ASTM A923-14 Method A	
Strumentazione Equipment		Metallographic microscope Leica DMI8 M (OAM n°1631)	
Parametri di prova Test parameters			
Posizione di prelievo Sampling position		Superficie interna Internal surface	
Ingrandimento Magnification		500X	
Caratteristiche provino Specimen information			
Preparazione della superficie Surface preparation	Lucidatura 1µm 1µm polishing	Attacco Etching	40% NaOH elettrolitico electrolytic
 			
Immagine non in scale / Image not in scale			
☒ Unaffected structure	The ferrite has been etched without revelationi of intermetallic phase The interphase boundaries are smooth		
☐ Possibly Affected Structure	The ferrite has been etched with isolated indications of possible intermetallic phase The interphase boundaries may show a fine waviness		
☐ Affected Structure	The indications of an intermetallic phase are readily revealed before or simultaneously with the staining of the ferrite during etching		
☐ Centerline Structure	An intermetallic phase is observed as a continuous or semi-continuous phase in the mid-thickness region of the product, with or without the affected structure outside of the mid-thickness region, indicative of segregation		
Note / Remarks L'analisi metallografica mostra una struttura regolare libera da fasi intermetalliche e precipitati The results of metallographic examination show a regular structure free from intermetallic phases and precipitates			

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	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	278287	LOTTO LOT	A1126

PROVA DI DETERMINAZIONE DELLE FASI INTERMETALLICHE DANNOSE / DETECTING DETRIMENTAL INTERMETALLIC PHASE TEST

Informazioni generali <i>General Info</i>			
Descrizione prova <i>Test description</i>		Prova di determ.delle fasi interm. dannose: Attacco con sodio idrossido <i>Detecting detrimental intermetallic phase: Sodium Hydroxide Etch Test</i>	
Standard Prova <i>Test standard</i>		ASTM A923-14 Metodo A ASTM A923-14 Method A	
Strumentazione <i>Equipment</i>		Metallographic microscope Leica DMi8 M (OAM n°1631)	
Parametri di prova <i>Test parameters</i>			
Posizione di prelievo <i>Sampling position</i>		Metà Spessore (T/2) <i>Mid Thickness (T/2)</i>	
Ingrandimento <i>Magnification</i>		500X	
Caratteristiche provino <i>Specimen information</i>			
Preparazione della superficie <i>Surface preparation</i>	Lucidatura 1µm 1µm polishing	Attacco Etching	40% NaOH elettrolitico electrolytic
 			
Immagine non in scale / Image not in scale			
☒ Unaffected structure	<i>The ferrite has been etched without revelationi of intermetallic phase The interphase boundaries are smooth</i>		
☐ Possibly Affected Structure	<i>The ferrite has been etched with isolated indications of possible intermetallic phase The interphase boundaries may show a fine waviness</i>		
☐ Affected Structure	<i>The indications of an intermetallic phase are readily revealed before or simultaneously with the staining of the ferrite during etching</i>		
☐ Centerline Structure	<i>An intermetallic phase is observed as a continuous or semi-continuous phase in the mid-thickness region of the product, with or without the affected structure outside of the mid-thickness region, indicative of segregation</i>		
Note / Remarks L'analisi metallografica mostra una struttura regolare libera da fasi intermetalliche e precipitati <i>The results of metallographic examination show a regular structure free from intermetallic phases and precipitates</i>			

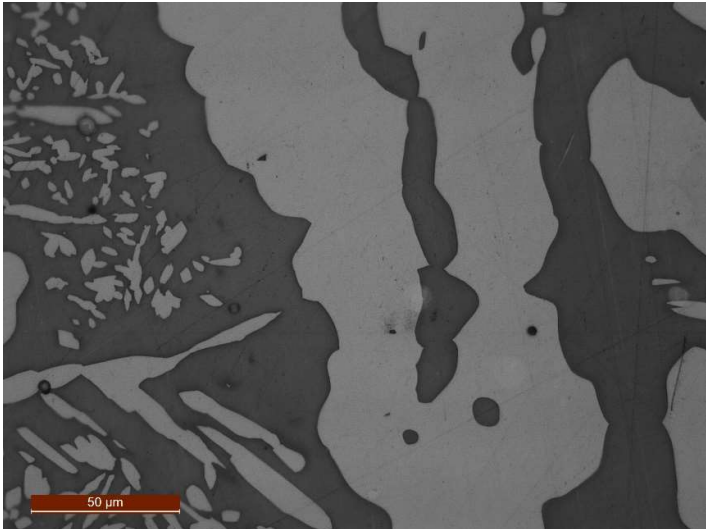
 23813 Cortenova (Lecco) ITALY Largo De Vecchi 11 P.o. box 11 Tel. 0341.98341 Fax 0341.983496 - 983497	DATA / DATE 04/04/2024		CERTIFICATO N° 0867/2024 REPORT N°	
	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	278287	LOTTO LOT	A1126

FRAZIONE DI VOLUME MEDIANTE SISTEMATICO CONTEGGIO MANUALE DI PUNTI / VOLUME FRACTION BY SYSTEMATIC MANUAL POINT COUNT

Informazioni generali General info			
Descrizione prova Test description		Frazione di vol. mediante sistematico conteggio manuale di punti Volume fraction by Systematic Manual Point Count	
Standard di prova Test standard		ASTM E562-19	
Strumentazione Equipment		Metallographic microscope Leica DMI8 M (OAM n°1631)	
Caratteristiche provino Specimen information			
Preparazione della superficie Surface preparation	Lucidatura 1µm 1µm polishing	Attacco Etching	20% NaOH elettrolitico electrolytic
Parametri di prova Test parameters			
		Ingrandimento Magnification	200X
N° campi X N° sezioni Nr fields X Nr sections	30 campi X 1 sezione 30 fields X 1 section	Tipo e dimensioni reticolo Grid type and size	Square grid, 40 x 40 mm 4 x 4 = 16 points
Risultati Results			
Posizione di Prelievo Sample Position		Superficie esterna External surface	
Vv :% Frazione di volume della ferrite ± 95% C.I. [%] % Volume fraction of the constituent ± 95% C.L.		49,25 ± 1,69	
Posizione di Prelievo Sample Position		Superficie interna Internal surface	
Vv :% Frazione di volume della ferrite ± 95% C.I. [%] % Volume fraction of the constituent ± 95% C.L.		50,12 ± 1,58	
Posizione di Prelievo Sample Position		Metà Spessore (T/2) Mid Thickness (T/2)	
Vv :% Frazione di volume della ferrite ± 95% C.I. [%] % Volume fraction of the constituent ± 95% C.L.		49,99 ± 1,85	
Valore Richiesto Required Value			
Vv% Minimo [%] Minimum	40	Vv% Massimo [%] Maximum	55
Note / Remarks			
			
Valore Richiesto : indicato nell'ordine cliente o nei documenti di riferimento / Required Value: indicated in the order or reference documents			

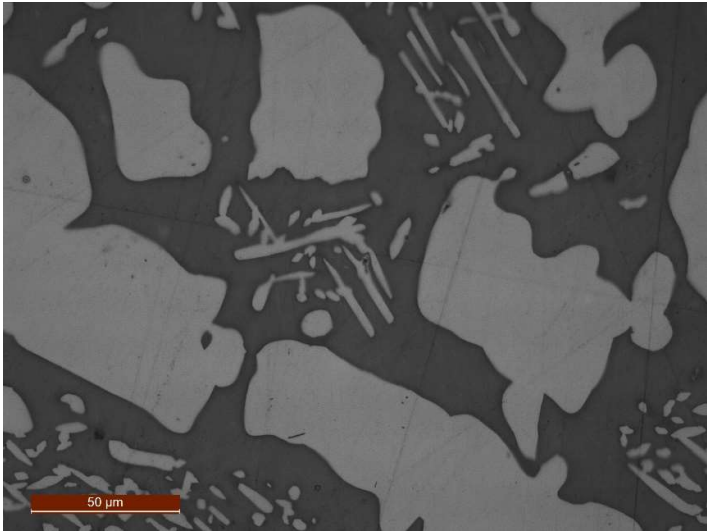
 23813 Cortenova (Lecco) ITALY Largo De Vecchi 11 P.o. box 11 Tel. 0341.98341 Fax 0341.983496 - 983497	DATA / DATE 04/04/2024		CERTIFICATO N° 0866/2024 REPORT N°	
	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	300309	LOTTO LOT	A1008

PROVA DI DETERMINAZIONE DELLE FASI INTERMETALLICHE DANNOSE / DETECTING DETRIMENTAL INTERMETALLIC PHASE TEST

Informazioni generali General Info			
Descrizione prova Test description		Prova di determ.delle fasi interm. dannose: Attacco con sodio idrossido Detecting detrimental intermetallic phase: Sodium Hydroxide Etch Test	
Standard Prova Test standard		ASTM A923-14 Metodo A ASTM A923-14 Method A	
Strumentazione Equipment		Metallographic microscope Leica DMi8 M (OAM n°1631)	
Parametri di prova Test parameters			
Posizione di prelievo Sampling position		Superficie esterna External surface	
Ingrandimento Magnification		500X	
Caratteristiche provino Specimen information			
Preparazione della superficie Surface preparation	Lucidatura 1µm 1µm polishing	Attacco Etching	40% NaOH elettrolitico electrolytic
 			
Immagine non in scale / Image not in scale			
☒ Unaffected structure	The ferrite has been etched without revelationi of intermetallic phase The interphase boundaries are smooth		
☐ Possibly Affected Structure	The ferrite has been etched with isolated indications of possible intermetallic phase The interphase boundaries may show a fine waviness		
☐ Affected Structure	The indications of an intermetallic phase are readily revealed before or simultaneously with the staining of the ferrite during etching		
☐ Centerline Structure	An intermetallic phase is observed as a continuous or semi-continuous phase in the mid-thickness region of the product, with or without the affected structure outside of the mid-thickness region, indicative of segregation		
Note / Remarks			
L'analisi metallografica mostra una struttura regolare libera da fasi intermetalliche e precipitati The results of metallographic examination show a regular structure free from intermetallic phases and precipitates			

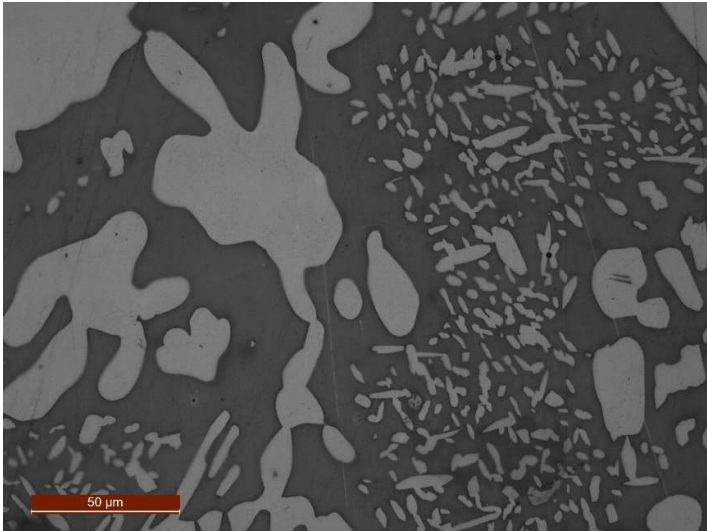
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	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	300309	LOTTO LOT	A1008

PROVA DI DETERMINAZIONE DELLE FASI INTERMETALLICHE DANNOSE / DETECTING DETRIMENTAL INTERMETALLIC PHASE TEST

Informazioni generali <i>General Info</i>			
Descrizione prova <i>Test description</i>		Prova di determ.delle fasi interm. dannose: Attacco con sodio idrossido <i>Detecting detrimental intermetallic phase: Sodium Hydroxide Etch Test</i>	
Standard Prova <i>Test standard</i>		ASTM A923-14 Metodo A ASTM A923-14 Method A	
Strumentazione <i>Equipment</i>		Metallographic microscope Leica DMi8 M (OAM n°1631)	
Parametri di prova <i>Test parameters</i>			
Posizione di prelievo <i>Sampling position</i>		Superficie interna <i>Internal surface</i>	
Ingrandimento <i>Magnification</i>		500X	
Caratteristiche provino <i>Specimen information</i>			
Preparazione della superficie <i>Surface preparation</i>	Lucidatura 1µm <i>1µm polishing</i>	Attacco <i>Etching</i>	40% NaOH elettrolitico <i>electrolytic</i>
 			
Immagine non in scale / Image not in scale			
☒ Unaffected structure	The ferrite has been etched without revelationi of intermetallic phase The interphase boundaries are smooth		
☐ Possibly Affected Structure	The ferrite has been etched with isolated indications of possible intermetallic phase The interphase boundaries may show a fine waviness		
☐ Affected Structure	The indications of an intermetallic phase are readily revealed before or simultaneously with the staining of the ferrite during etching		
☐ Centerline Structure	An intermetallic phase is observed as a continuous or semi-continuous phase in the mid-thickness region of the product, with or without the affected structure outside of the mid-thickness region, indicative of segregation		
Note / Remarks			
L'analisi metallografica mostra una struttura regolare libera da fasi intermetalliche e precipitati <i>The results of metallographic examination show a regular structure free from intermetallic phases and precipitates</i>			

 23813 Cortenova (Lecco) ITALY Largo De Vecchi 11 P.o. box 11 Tel. 0341.98341 Fax 0341.983496 - 983497	DATA / DATE 04/04/2024		CERTIFICATO N° 0866/2024 REPORT N°	
	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	300309	LOTTO LOT	A1008

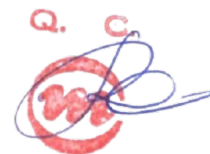
PROVA DI DETERMINAZIONE DELLE FASI INTERMETALLICHE DANNOSE / DETECTING DETRIMENTAL INTERMETALLIC PHASE TEST

Informazioni generali <i>General Info</i>			
Descrizione prova <i>Test description</i>		Prova di determ.delle fasi interm. dannose: Attacco con sodio idrossido <i>Detecting detrimental intermetallic phase: Sodium Hydroxide Etch Test</i>	
Standard Prova <i>Test standard</i>		ASTM A923-14 Metodo A ASTM A923-14 Method A	
Strumentazione <i>Equipment</i>		Metallographic microscope Leica DMi8 M (OAM n°1631)	
Parametri di prova <i>Test parameters</i>			
Posizione di prelievo <i>Sampling position</i>		Metà Spessore (T/2) <i>Mid Thickness (T/2)</i>	
Ingrandimento <i>Magnification</i>		500X	
Caratteristiche provino <i>Specimen information</i>			
Preparazione della superficie <i>Surface preparation</i>	Lucidatura 1µm 1µm polishing	Attacco Etching	40% NaOH elettrolitico electrolytic
			
Immagine non in scale / Image not in scale			
☒ Unaffected structure	<i>The ferrite has been etched without revelationi of intermetallic phase The interphase boundaries are smooth</i>		
☐ Possibly Affected Structure	<i>The ferrite has been etched with isolated indications of possible intermetallic phase The interphase boundaries may show a fine waviness</i>		
☐ Affected Structure	<i>The indications of an intermetallic phase are readily revealed before or simultaneously with the staining of the ferrite during etching</i>		
☐ Centerline Structure	<i>An intermetallic phase is observed as a continuous or semi-continuous phase in the mid-thickness region of the product, with or without the affected structure outside of the mid-thickness region, indicative of segregation</i>		
Note / Remarks L'analisi metallografica mostra una struttura regolare libera da fasi intermetalliche e precipitati <i>The results of metallographic examination show a regular structure free from intermetallic phases and precipitates</i>			

 23813 Cortenova (Lecco) ITALY Largo De Vecchi 11 P.o. box 11 Tel. 0341.98341 Fax 0341.983496 - 983497	DATA / DATE 04/04/2024		CERTIFICATO N° 0866/2024 REPORT N°	
	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	300309	LOTTO LOT	A1008

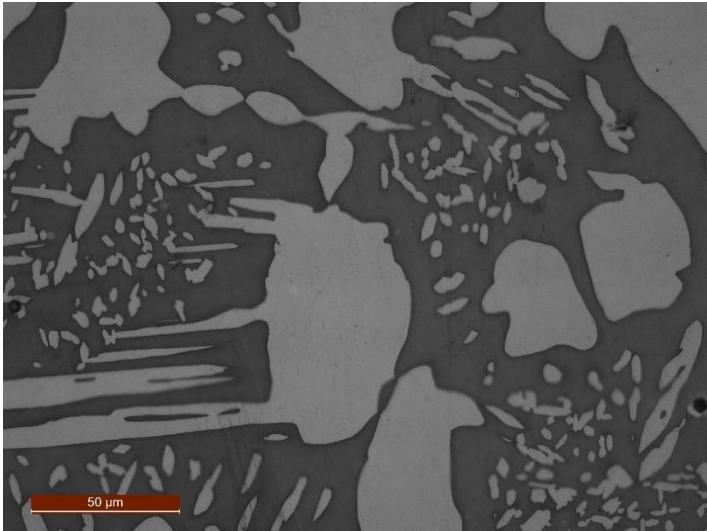
FRAZIONE DI VOLUME MEDIANTE SISTEMATICO CONTEGGIO MANUALE DI PUNTI / VOLUME FRACTION BY SYSTEMATIC MANUAL POINT COUNT

Informazioni generali <i>General info</i>			
Descrizione prova <i>Test description</i>		Frazione di vol. mediante sistematico conteggio manuale di punti <i>Volume fraction by Systematic Manual Point Count</i>	
Standard di prova <i>Test standard</i>		ASTM E562-19	
Strumentazione <i>Equipment</i>		Metallographic microscope Leica DMI8 M (OAM n°1631)	
Caratteristiche provino <i>Specimen information</i>			
Preparazione della superficie <i>Surface preparation</i>	Lucidatura 1µm 1µm polishing	Attacco Etching	20% NaOH elettrolitico electrolytic
Parametri di prova <i>Test parameters</i>			
		Ingrandimento Magnification	200X
N° campi X N° sezioni Nr fields X Nr sections	30 campi X 1 sezione 30 fields X 1 section	Tipo e dimensioni reticolo Grid type and size	Square grid, 40 x 40 mm 4 x 4 = 16 points
Risultati <i>Results</i>			
Posizione di Prelievo Sample Position		Superficie esterna External surface	
Vv :% Frazione di volume della ferrite ± 95% C.I. [%] % Volume fraction of the constituent ± 95% C.L.		46,08 ± 2,04	
Posizione di Prelievo Sample Position		Superficie interna Internal surface	
Vv :% Frazione di volume della ferrite ± 95% C.I. [%] % Volume fraction of the constituent ± 95% C.L.		47,77 ± 1,36	
Posizione di Prelievo Sample Position		Metà Spessore (T/2) Mid Thickness (T/2)	
Vv :% Frazione di volume della ferrite ± 95% C.I. [%] % Volume fraction of the constituent ± 95% C.L.		48,01 ± 1,44	
Valore Richiesto Required Value			
Vv% Minimo [%] Minimum	40	Vv% Massimo [%] Maximum	55
Note / Remarks			
Valore Richiesto : indicato nell'ordine cliente o nei documenti di riferimento / Required Value: indicated in the order or reference documents			



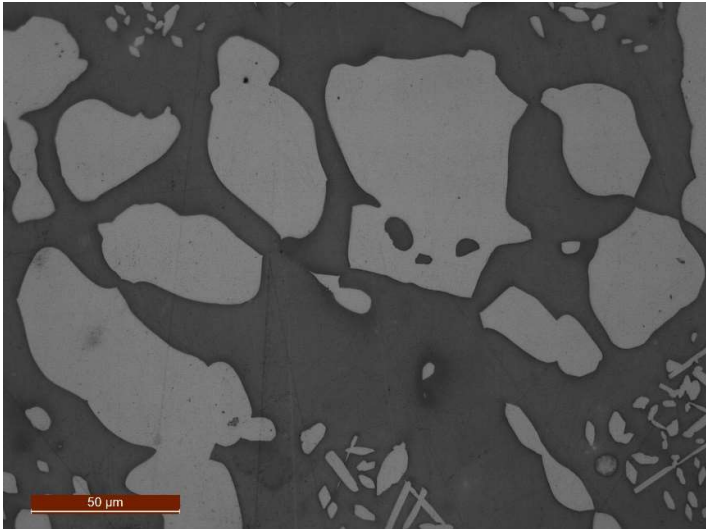
 23813 Cortenova (Lecco) ITALY Largo De Vecchi 11 P.o. box 11 Tel. 0341.98341 Fax 0341.983496 - 983497	DATA / DATE 04/04/2024		CERTIFICATO N° 0868/2024 REPORT N°	
	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	377897	LOTTO LOT	A1261

PROVA DI DETERMINAZIONE DELLE FASI INTERMETALLICHE DANNOSE / DETECTING DETRIMENTAL INTERMETALLIC PHASE TEST

Informazioni generali General Info			
Descrizione prova Test description		Prova di determ.delle fasi interm. dannose: Attacco con sodio idrossido Detecting detrimental intermetallic phase: Sodium Hydroxide Etch Test	
Standard Prova Test standard		ASTM A923-14 Metodo A ASTM A923-14 Method A	
Strumentazione Equipment		Metallographic microscope Leica DMi8 M (OAM n°1631)	
Parametri di prova Test parameters			
Posizione di prelievo Sampling position		Superficie esterna External surface	
Ingrandimento Magnification		500X	
Caratteristiche provino Specimen information			
Preparazione della superficie Surface preparation	Lucidatura 1µm 1µm polishing	Attacco Etching	40% NaOH elettrolitico electrolytic
 			
Immagine non in scale / Image not in scale			
☒ Unaffected structure	The ferrite has been etched without revelationi of intermetallic phase The interphase boundaries are smooth		
☐ Possibly Affected Structure	The ferrite has been etched with isolated indications of possible intermetallic phase The interphase boundaries may show a fine waviness		
☐ Affected Structure	The indications of an intermetallic phase are readily revealed before or simultaneously with the staining of the ferrite during etching		
☐ Centerline Structure	An intermetallic phase is observed as a continuous or semi-continuous phase in the mid-thickness region of the product, with or without the affected structure outside of the mid-thickness region, indicative of segregation		
Note / Remarks			
L'analisi metallografica mostra una struttura regolare libera da fasi intermetalliche e precipitati The results of metallographic examination show a regular structure free from intermetallic phases and precipitates			

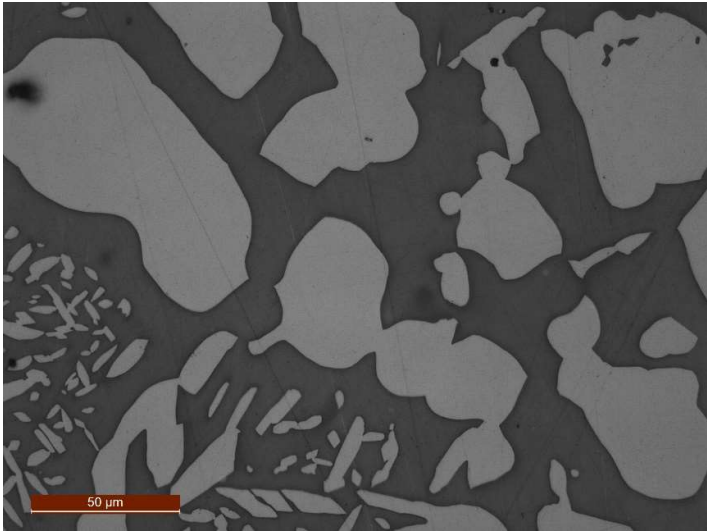
 23813 Cortenova (Lecco) ITALY Largo De Vecchi 11 P.o. box 11 Tel. 0341.98341 Fax 0341.983496 - 983497	DATA / DATE 04/04/2024		CERTIFICATO N° 0868/2024 REPORT N°	
	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	377897	LOTTO LOT	A1261

PROVA DI DETERMINAZIONE DELLE FASI INTERMETALLICHE DANNOSE / DETECTING DETRIMENTAL INTERMETALLIC PHASE TEST

Informazioni generali General Info			
Descrizione prova Test description		Prova di determ.delle fasi interm. dannose: Attacco con sodio idrossido Detecting detrimental intermetallic phase: Sodium Hydroxide Etch Test	
Standard Prova Test standard		ASTM A923-14 Metodo A ASTM A923-14 Method A	
Strumentazione Equipment		Metallographic microscope Leica DMI8 M (OAM n°1631)	
Parametri di prova Test parameters			
Posizione di prelievo Sampling position		Superficie interna Internal surface	
Ingrandimento Magnification		500X	
Caratteristiche provino Specimen information			
Preparazione della superficie Surface preparation	Lucidatura 1µm 1µm polishing	Attacco Etching	40% NaOH elettrolitico electrolytic
 			
Immagine non in scale / Image not in scale			
☒ Unaffected structure	The ferrite has been etched without revelationi of intermetallic phase The interphase boundaries are smooth		
☐ Possibly Affected Structure	The ferrite has been etched with isolated indications of possible intermetallic phase The interphase boundaries may show a fine waviness		
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Note / Remarks			
L'analisi metallografica mostra una struttura regolare libera da fasi intermetalliche e precipitati The results of metallographic examination show a regular structure free from intermetallic phases and precipitates			

 23813 Cortenova (Lecco) ITALY Largo De Vecchi 11 P.o. box 11 Tel. 0341.98341 Fax 0341.983496 - 983497	DATA / DATE 04/04/2024		CERTIFICATO N° 0868/2024 REPORT N°	
	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	377897	LOTTO LOT	A1261

PROVA DI DETERMINAZIONE DELLE FASI INTERMETALLICHE DANNOSE / DETECTING DETRIMENTAL INTERMETALLIC PHASE TEST

Informazioni generali General Info			
Descrizione prova Test description		Prova di determ.delle fasi interm. dannose: Attacco con sodio idrossido Detecting detrimental intermetallic phase: Sodium Hydroxide Etch Test	
Standard Prova Test standard		ASTM A923-14 Metodo A ASTM A923-14 Method A	
Strumentazione Equipment		Metallographic microscope Leica DMI8 M (OAM n°1631)	
Parametri di prova Test parameters			
Posizione di prelievo Sampling position		Metà Spessore (T/2) Mid Thickness (T/2)	
Ingrandimento Magnification		500X	
Caratteristiche provino Specimen information			
Preparazione della superficie Surface preparation	Lucidatura 1µm 1µm polishing	Attacco Etching	40% NaOH elettrolitico electrolytic
 			
Immagine non in scale / Image not in scale			
☒ Unaffected structure	The ferrite has been etched without revelationi of intermetallic phase The interphase boundaries are smooth		
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☐ Affected Structure	The indications of an intermetallic phase are readily revealed before or simultaneously with the staining of the ferrite during etching		
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Note / Remarks			
L'analisi metallografica mostra una struttura regolare libera da fasi intermetalliche e precipitati The results of metallographic examination show a regular structure free from intermetallic phases and precipitates			

 23813 Cortenova (Lecco) ITALY Largo De Vecchi 11 P.o. box 11 Tel. 0341.98341 Fax 0341.983496 - 983497	DATA / DATE 04/04/2024		CERTIFICATO N° 0868/2024 REPORT N°	
	MATERIALE MATERIAL		ASTM A182 F51/F60	
	COLATA HEAT N°	377897	LOTTO LOT	A1261

FRAZIONE DI VOLUME MEDIANTE SISTEMATICO CONTEGGIO MANUALE DI PUNTI / VOLUME FRACTION BY SYSTEMATIC MANUAL POINT COUNT

Informazioni generali <i>General info</i>			
Descrizione prova <i>Test description</i>		Frazione di vol. mediante sistematico conteggio manuale di punti <i>Volume fraction by Systematic Manual Point Count</i>	
Standard di prova <i>Test standard</i>		ASTM E562-19	
Strumentazione <i>Equipment</i>		Metallographic microscope Leica DMI8 M (OAM n°1631)	
Caratteristiche provino <i>Specimen information</i>			
Preparazione della superficie <i>Surface preparation</i>	Lucidatura 1µm 1µm polishing	Attacco Etching	20% NaOH elettrolitico electrolytic
Parametri di prova <i>Test parameters</i>			
		Ingrandimento Magnification	200X
N° campi X N° sezioni <i>Nr fields X Nr sections</i>	30 campi X 1 sezione 30 fields X 1 section	Tipo e dimensioni reticolo Grid type and size	Square grid, 40 x 40 mm 4 x 4 = 16 points
Risultati <i>Results</i>			
Posizione di Prelievo Sample Position		Superficie esterna External surface	
Vv :% Frazione di volume della ferrite ± 95% C.I. [%] % Volume fraction of the constituent ± 95% C.L.		46,55 ± 1,30	
Posizione di Prelievo Sample Position		Superficie interna Internal surface	
Vv :% Frazione di volume della ferrite ± 95% C.I. [%] % Volume fraction of the constituent ± 95% C.L.		46,98 ± 1,42	
Posizione di Prelievo Sample Position		Metà Spessore (T/2) Mid Thickness (T/2)	
Vv :% Frazione di volume della ferrite ± 95% C.I. [%] % Volume fraction of the constituent ± 95% C.L.		47,23 ± 1,03	
Valore Richiesto Required Value			
Vv% Minimo [%] Minimum	40	Vv% Massimo [%] Maximum	55
Note / Remarks			
Valore Richiesto : indicato nell'ordine cliente o nei documenti di riferimento / Required Value: indicated in the order or reference documents			

Q. C.





OFFICINE AMBROGIO MELESI & C. S.R.L.

Largo De Vecchi 11 Phone: +39.0341.98341 Cap.Soc./Share Cap. €500.000 i.v.
23813 Cortenova (LC) Fax: +39.0341.983496 / 7 P.IVA - C.F. - R.I. Lecco:
Italy E-mail: info@melesi.it IT 00231630138
Web site: www.melesi.it R.E.A. n. LC-108835
N.Mecc./Export n. LC005895



Hardness Test Report

HRD_191621

Customer FRAMO FLATOY AS Melesi Job 75698

P.O. 1002-PO-013388

Measurement on BODY

Number of Measurements 2

Item	Q.ty	Description	Heat Number	Area Tested	Result [HB]
20	2	F51/F60 PREMACHINED RING	278287	BODY	223 - 223 - 217 - 216
60	2	F51/F60 PREMACHINED RING	377897	BODY	221 - 226 - 210 - 226
40	2	F51/F60 PREMACHINED RING	300309	BODY	223 - 221 - 214 - 220

Date 04/04/2024





SR 1852

COGNE ACCIAI SPECIALI S.p.a.

ITALY - 11100 AOSTA - VIA PARAVERA, 16
TEL +39.0165.3021 - FAX +39.0165.302296
CAP. SOC. 250.000.000 EUR INT. VERS.
C.F. 02187360967
VAT: IT00571320076 - R.E.A. n. AO-50474

Company with Management Systems
certified according to ISO 9001, ITAF 16949,
EN 9100, ISO 14001 and ISO 45001.

INSPECTION CERTIFICATE 3.1 (EN 10204:2004)
DOCUMENT NUMBER 2022049752
PAGE 1/1

CUSTOMER:
CUSTOMER'S ORDER:
MANUFACTURER'S WORKS:
PRODUCER OF THE DOC:
MANUFACTURER'S WORKS ORDER NO:
MARK OF THE MANUFACTURER:
DISPATCH NOTE:

OFFICINE AMBROGIO MELESI & C. SRL
21/03419
AOSTA, VIA PARAVERA 16 - ITALY
QUALITY DEPARTMENT
0025298193 / 000010
COGNE
0081514452 / 000010 WEIGHT: 19000 KG / 41895 LB

INTERNAL SPEC:
PRODUCT:
SURFACE FINISH:
PRODUCT DELIVERY CONDITION:
PRODUCT DIMENSIONS (mm):
GRADE:
IDENTIFICATION HEAT NO:
MARKING OF THE PRODUCT:

MELESIF51A182
COL CAST FLAT BLOOMS TOLERANCE: + 0 - 0 (+ 0 - 0 in)
GR ROUGH
220,000 270,000 (8,661 10,630 in) LENGTH (mm) 04000 04100 (00157 00161 in)
F51/F60 INTERNAL GRADE: 329A
278287 IDENTIFICATION LOT NO: 861940
F51

STEELMAKING EAF + AOD + CONTINUOUS CASTING

CHEMICAL LADLE ANALYSIS ACCORDING TO ASTM A751 - EN ISO 14284

Control lot No. - Weight	83.888 KG -				184.938 LB							
ELEMENTS	C	Si	Mn	P	S	N	Al	Cr	Mo	Ni		
MEASUREMENT UNIT	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w		
OBTAINED	0,014	0,39	1,18	0,026	0,0009	0,18	0,006	22,47	3,24	5,71		
ELEMENTS	Cu	Sn	Co	B	CV_F8							
MEASUREMENT UNIT	% w/w	% w/w	% w/w	% w/w	% w/w							
OBTAINED	0,26	0,006	0,13	0,0023	36							
CV_F8 =	CR+3.3*MO+16*N2											

WE GUARANTEE FERRITE CONTENTS (ACCORDING TO ASTM E562/02) BETWEEN
45% AND 55%

ULTRASONIC TEST AT 100% : SATISFACTORY

RADIOACTIVITY CONTROL PERFORMED WITH A GAMMA SPECTROMETER: ASTM E181.

GAMMA ACTIVITY ON THE PRODUCTS BELOW 0.1 BQ/G ACCORDING TO EU RESOLUTION 2013/59/EURATOM , IAEA RS-G-1.7 AND IAEA SSG-17.

MATERIAL FREE FROM MERCURY CONTAMINATION

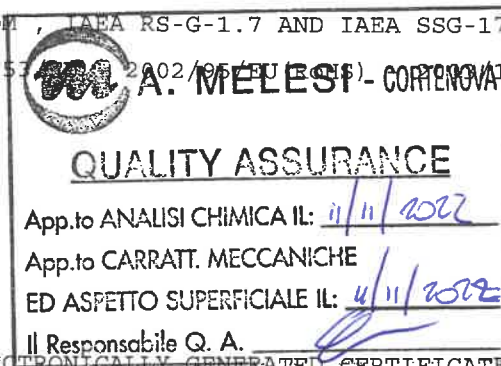
NO WELDING REPAIR ON THE MATERIAL.

RADIOACTIVITY CONTROL PERFORMED WITH A GAMMA SPECTROMETER: ASTM E181.

GAMMA ACTIVITY ON THE PRODUCTS BELOW 0.1 BQ/G ACCORDING TO EU RESOLUTION 2013/59/EURATOM , IAEA RS-G-1.7 AND IAEA SSG-17.

SAWABLE MATERIAL.

THE PRODUCT SATISFIES THE DIRECTIVES : 2015/863/EU (ROHS3) - 2011/65/EU (ROHS2) - 2000/531/EC (ROHS) - 2002/95/EU (RoHS) - 2003/11/EU - 2005/618/EU.



DATE 31.10.2022

BARBARA VIETTI MV(QUALITY CERTIFICATOR) - ELECTRONICALLY GENERATED CERTIFICATE

-Cogne



SR1870

COGNE ACCIAI SPECIALI S.p.a.

ITALY - 11100 AOSTA - VIA PARAVERA, 16
TEL +39.0165.3021 - FAX +39.0165.302296
CAP. SOC. 250.000.000 EUR INT. VERS.
C.F. 02187360967
VAT: IT00571320076 - R.E.A. n. AO-50474

Company with Management Systems
certified according to ISO 9001, ITAF 16949,
EN 9100, ISO 14001 and ISO 45001.

INSPECTION CERTIFICATE 3.1 (EN 10204:2004)
DOCUMENT NUMBER 2023008195
PAGE 1/1

CUSTOMER:
CUSTOMER'S ORDER:
MANUFACTURER'S WORKS:
PRODUCER OF THE DOC:
MANUFACTURER'S WORKS ORDER NO:
MARK OF THE MANUFACTURER:
DISPATCH NOTE:

OFFICINE AMBROGIO MELESI & C. SRL
21/03460
AOSTA, VIA PARAVERA 16 - ITALY
QUALITY DEPARTMENT
0025300349 / 000030
COGNE
0081536209 / 000010 WEIGHT: 5310 KG / 11709 LB

INTERNAL SPEC:
PRODUCT:
SURFACE FINISH:
PRODUCT DELIVERY CONDITION:
PRODUCT DIMENSIONS (mm):
GRADE:
IDENTIFICATION HEAT NO:
MARKING OF THE PRODUCT:

MELESIF51A182
FUC FORGED round 500 TOLERANCE: + 15,000 - 15,000 (+ 0,591 - 0,591 in)
GR ROUGH
300,000 (11,811 in) LENGTH (mm) 04000 07500 (00157 0029 in)
F51/F60 INTERNAL GRADE: 329A
300309 IDENTIFICATION LOT NO: 036210
F51

STEELMAKING EAF + AOD
REDUCTION RATIO 2,6

CHEMICAL LADLE ANALYSIS ACCORDING TO ASTM A751 - EN ISO 14284

Control lot No. - Weight :020000957379 -		78.737 KG -		173.582 LB											
ELEMENTS	C	Si	Mn	P	S	N	Al	Cr	Mo	Ni					
MEASUREMENT UNIT	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w					
OBTAINED	0,021	0,44	1,19	0,020	0,0005	0,18	0,011	22,46	3,20	5,76					
ELEMENTS	Cu	Sn	Co	B	CV_F8										
MEASUREMENT UNIT	% w/w	% w/w	% w/w	% w/w	% w/w										
OBTAINED	0,13	0,004	0,11	0,0029	36										
CV_F8 =	CR+3.3*MO+16*N2														

WE GUARANTEE FERRITE CONTENTS (ACCORDING TO ASTM E562/02) BETWEEN
45% AND 55%

ULTRASONIC TEST AT 100% : SATISFACTORY

RADIOACTIVITY CONTROL PERFORMED WITH A GAMMA SPECTROMETER: ASTM E181.

GAMMA ACTIVITY ON THE PRODUCTS BELOW 0.1 BQ/G ACCORDING TO EU RESOLUTION 2013/59/EURATOM , IAEA RS-G-1.7 AND IAEA SSG-17.

MATERIAL FREE FROM MERCURY CONTAMINATION

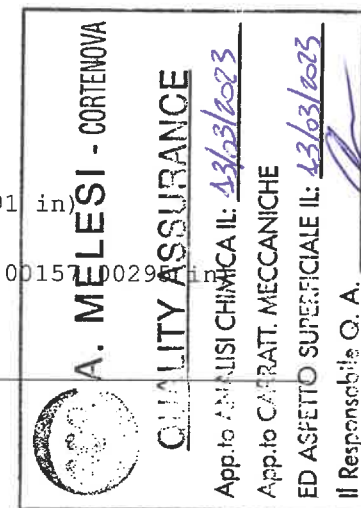
NO WELDING REPAIR ON THE MATERIAL.

RADIOACTIVITY CONTROL PERFORMED WITH A GAMMA SPECTROMETER: ASTM E181.

GAMMA ACTIVITY ON THE PRODUCTS BELOW 0.1 BQ/G ACCORDING TO EU RESOLUTION 2013/59/EURATOM , IAEA RS-G-1.7 AND IAEA SSG-17.

SAWABLE MATERIAL.

THE PRODUCT SATISFIES THE DIRECTIVES : 2015/863/EU (ROHS3) - 2011/65/EU (ROHS2) - 2000/53/EU - 2002/95/EU (RoHS) - 2003/11/EU - 2005/618/EU.





COGNE ACCIAI SPECIALI S.p.a.

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C.A.P. SOC. 905563 - 167,00 EUR INT. VERS.
C.F. 02187360987
VAT: IT00571320076 - R.E.A. n. AO-50474

Company with Management Systems
certified according to ISO 9001, ITAF 16949,
EN 9100, ISO 14001 and ISO 45001.

CUSTOMER:
CUSTOMER'S ORDER:
MANUFACTURER'S WORKS:
PRODUCER OF THE DOC:
MANUFACTURER'S WORKS ORDER NO:
MARK OF THE MANUFACTURER:
DISPATCH NOTE:

OFFICINE AMBROGIO MELESI & C. SRL
21/03589
AOSTA, VIA PARAVERA 16 - ITALY
QUALITY DEPARTMENT
0025307876 / 000020
COGNE
0081578718 / 000010 WEIGHT: 22730 KG / 50111 LB

INTERNAL SPEC:

PRODUCT:

SURFACE FINISH:

PRODUCT DELIVERY CONDITION:

PRODUCT DIMENSIONS (mm):

GRADE:

IDENTIFICATION HEAT NO:

MARKING OF THE PRODUCT:

MELESIP51A182

COL CAST FLAT BLOOMS TOLERANCE: + 0 - 0 (+ 0 - 0 in)

GR ROUGH

220.000 270.000 (8,661 10,630 in) LENGTH (mm)

F51/F60

377897

F51

04000 04100 (00157 00161 in)

329A

436390

INTERNAL GRADE:

IDENTIFICATION LOT NO:

STEELMAKING EAF + AOD + CONTINUOUS CASTING

CHEMICAL LADLE ANALYSIS ACCORDING TO ASTM A751 - EN ISO 14284

Control lot No. - Weight : 020000987188 -

ELEMENTS	C	Si	Mn	P	S	N	Al	Cr	Mo	Ni
MEASUREMENT UNIT	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w	% w/w
OBTAINED	0,024	0,37	1,21	0,025	0,0005	0,18	0,006	22,46	3,17	5,62
ELEMENTS	Cu	Sn	Co	B	CV F8					
MEASUREMENT UNIT	% w/w	% w/w	% w/w	% w/w	% w/w					
OBTAINED	0,21	0,005	0,14	0,0022	36					
CV_F8 =	CR+3.3*MO+16*N2									

WE GUARANTEE FERRITE CONTENTS (ACCORDING TO ASTM E562/02) BETWEEN

45% AND 55%

ULTRASONIC TEST AT 100% : SATISFACTORY

RADIOACTIVITY CONTROL PERFORMED WITH A GAMMA SPECTROMETER: ASTM E181.

GAMMA ACTIVITY ON THE PRODUCTS BELOW 0.1 BQ/G ACCORDING TO EU RESOLUTION 2013/59/EURATOM , IAEA RS-G-1.7 AND IAEA SSG-17.

MATERIAL FREE FROM MERCURY CONTAMINATION

NO WELDING REPAIR ON THE MATERIAL.

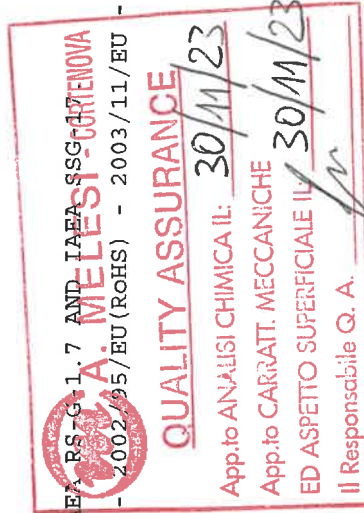
RADIOACTIVITY CONTROL PERFORMED WITH A GAMMA SPECTROMETER: ASTM E181.

GAMMA ACTIVITY ON THE PRODUCTS BELOW 0.1 BQ/G ACCORDING TO EU RESOLUTION 2013/59/EURATOM , IAEA RS-G-1.7 AND IAEA SSG-17.

SAWABLE MATERIAL.

THE PRODUCT SATISFIES THE DIRECTIVES : 2015/863/EU (ROHS3) - 2011/65/EU (ROHS2) - 2000/53/EU - 2003/11/EU

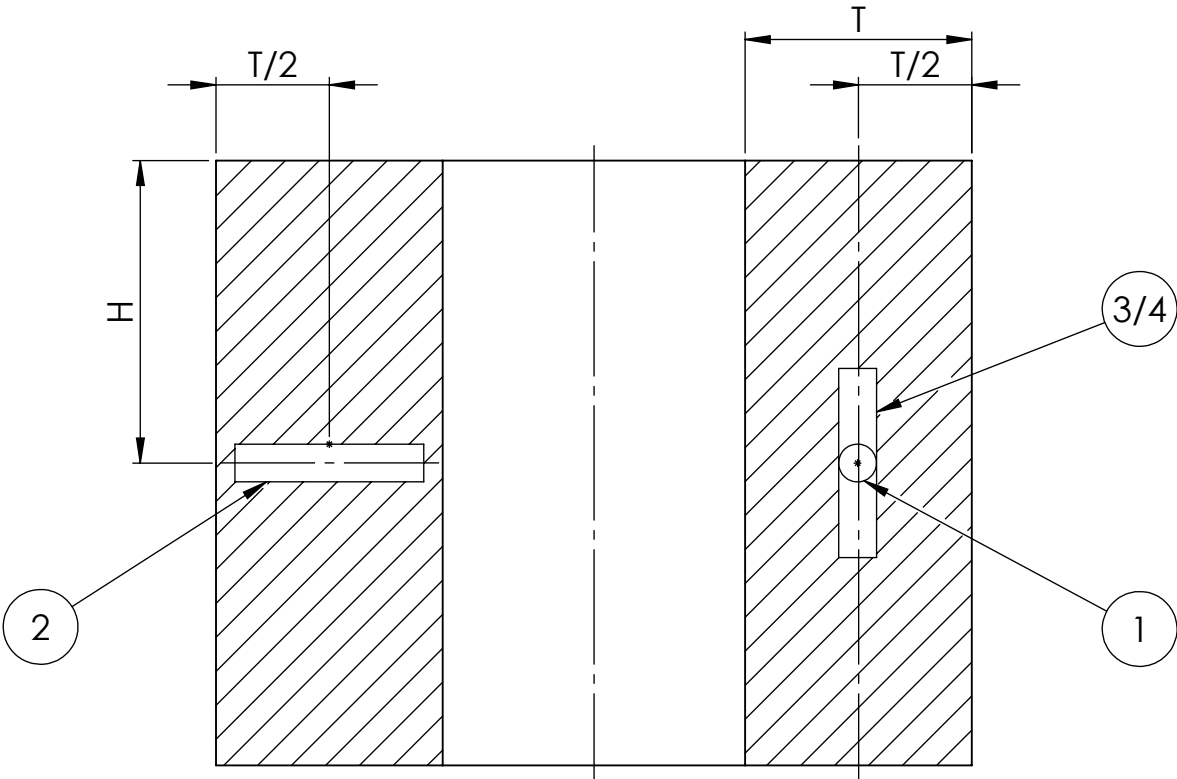
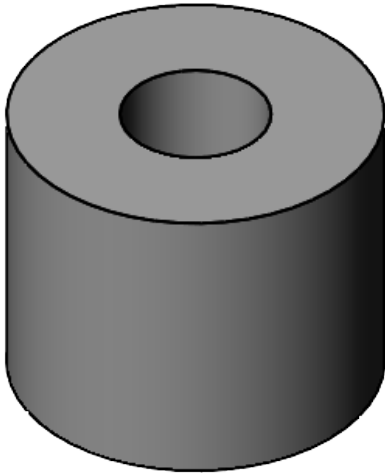
2005/618/EU.



DATE 22.11.2023

MORABITO ROSA CF (QUALITY CERTIFICATOR) - ELECTRONICALLY GENERATED CERTIFICATE

-Cogne

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E		Posizioni di prelievo: 1) Direzione trasversale, a T/2 2) Direzione radiale, a T/2 3/4) Direzione longitudinale, a T/2 Test location: 1) Trasversal direction, at T/2 2) Radial direction 3/4) Longitudinal direction, at T/2							E																																																																																																								
F	*Profondità di estrazione H in funzione della specifica di riferimento. <table><tr><td colspan="6">Not in scale.</td><td colspan="2">OFFICINE AMBROGIO MELESI</td></tr><tr><td colspan="6">Reference drawing : -</td><td colspan="2">TITLE: TEST LOCATION - ANELLO A SEZIONE RETTANGOLARE</td></tr><tr><td colspan="6">The whole or any part of this drawing cannot be reproduced and/or transferred to any third part without our written permission</td><td colspan="2">JOB NR - POS. NR</td></tr><tr><td colspan="6"></td><td colspan="2">-</td></tr><tr><td colspan="6"></td><td colspan="2">MATERIAL:</td></tr><tr><td colspan="6">0 ISSUED 10/01/2024 M.Selva M.Mogliuzzi A.Panzeri</td><td colspan="2">WEIGHT [kg]:</td></tr><tr><td colspan="6">Rev Description Date Drawn Checked Approved</td><td colspan="2">- -</td></tr><tr><td colspan="6"></td><td colspan="2">CUSTOMER :</td></tr><tr><td colspan="6"></td><td colspan="2">PROJECT : -</td></tr><tr><td colspan="6"></td><td colspan="2">PO NO : -</td></tr><tr><td colspan="6"></td><td colspan="2">TAG No. : -</td></tr><tr><td colspan="6"></td><td colspan="2">ITEM No. : - QTY: -</td></tr><tr><td colspan="6">1 2 3 4</td><td colspan="2">A3 Scale : NTS  Sheet 1 of 1</td></tr></table>								Not in scale.						OFFICINE AMBROGIO MELESI		Reference drawing : -						TITLE: TEST LOCATION - ANELLO A SEZIONE RETTANGOLARE		The whole or any part of this drawing cannot be reproduced and/or transferred to any third part without our written permission						JOB NR - POS. NR								-								MATERIAL:		0 ISSUED 10/01/2024 M.Selva M.Mogliuzzi A.Panzeri						WEIGHT [kg]:		Rev Description Date Drawn Checked Approved						- -								CUSTOMER :								PROJECT : -								PO NO : -								TAG No. : -								ITEM No. : - QTY: -		1 2 3 4						A3 Scale : NTS  Sheet 1 of 1		F
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